

1 Solution — GIAM 6.6

Problem 2

1.1 Problem

The usual algebraic procedure for inverting

$$T(x) = \frac{x^2 + x}{2}$$

stalls — solving for y in $y = T(x)$ is not a chain of inverse operations but a quadratic solve. Use the *geometry* of functions and inverses to find a formula for T^{-1} . (Hint: invert the simpler $S(x) = x^2/2$ first; that pins down the right vertical scaling factor.)

1.2 Why the “Usual Algebra” Stalls

The “swap x and y , then isolate y ” recipe gives

$$y = \frac{x^2 + x}{2} \xrightarrow{\text{swap}} x = \frac{y^2 + y}{2} \implies y^2 + y - 2x = 0.$$

This is quadratic in y , so isolating y requires the quadratic formula and a $\pm$ choice — not a clean undo-by-inverse-operations chain. Functional, but unsatisfying.

1.3 Geometric Reframe — T Is a Translated Copy of S

Complete the square:

$$T(x) = \frac{x^2 + x}{2} = \frac{(x + \frac{1}{2})^2 - \frac{1}{4}}{2} = \frac{(x + \frac{1}{2})^2}{2} - \frac{1}{8}.$$

Defining $S(x) = x^2/2$, this reads

$$T(x) = S(x + \tfrac{1}{2}) - \tfrac{1}{8}.$$

So T is S translated left by $\frac{1}{2}$ and down by $\frac{1}{8}$.

1.4 Step 1 — Invert S

For $S(x) = x^2/2$ on the restricted domain $x \geq 0$,

$$y = \frac{x^2}{2} \implies x^2 = 2y \implies x = \sqrt{2y},$$

so

$$S^{-1}(y) = \sqrt{2y}.$$

The factor of **2** inside the radical is the *reciprocal* of the $\frac{1}{2}$ scaling outside the square in S . This is the “right vertical scaling factor” the hint pointed at: vertical compression by $\frac{1}{2}$ in S becomes horizontal stretching by **2** inside the radical of S^{-1} .

1.5 Step 2 — The Translation-Swap Rule

Reflection across the line $y = x$ — which is the geometric content of inverting a function — swaps the x - and y -axes. Consequently, **horizontal and vertical translations swap under inversion.**

Concretely, if $g(x) = f(x + a) + b$, then

$$g^{-1}(y) = f^{-1}(y - b) - a.$$

The input shift $+a$ (translate left a) on g becomes an output shift $-a$ (translate down a) on g^{-1} ; the output shift $+b$ (translate up b) on g becomes an input shift $-b$ (translate right b) on g^{-1} .

1.6 Step 3 — Combine

For $T = S(x + \frac{1}{2}) - \frac{1}{8}$: take $a = \frac{1}{2}$ and $b = -\frac{1}{8}$. The translation-swap rule gives

$$T^{-1}(y) = S^{-1}(y - (-\frac{1}{8})) - \frac{1}{2} = S^{-1}(y + \frac{1}{8}) - \frac{1}{2}.$$

Plugging in $S^{-1}(y) = \sqrt{2y}$,

$$T^{-1}(y) = \sqrt{2y + \frac{1}{4}} - \frac{1}{2}, \quad y \geq -\frac{1}{8}.$$

1.7 Verification — Same as Problem 1

This must match the quadratic-formula answer from Problem 1, $T^{-1}(y) = \frac{-1 + \sqrt{1 + 8y}}{2}$. A bit of algebra:

$$\sqrt{2y + \frac{1}{4}} - \frac{1}{2} = \sqrt{\frac{8y+1}{4}} - \frac{1}{2} = \frac{\sqrt{8y+1}}{2} - \frac{1}{2} = \frac{-1 + \sqrt{1 + 8y}}{2}. \quad \checkmark$$

The geometric and algebraic routes converge on the same formula — which is reassuring, and shows the geometry was doing real work, not a magic trick.

1.8 Sanity Check

A few values of T^{-1} at triangular-number inputs:

y	$T^{-1}(y) = \sqrt{2y + \frac{1}{4}} - \frac{1}{2}$	check
0	$\sqrt{\frac{1}{4}} - \frac{1}{2} = 0$	$T(0) = 0$
1	$\sqrt{\frac{9}{4}} - \frac{1}{2} = 1$	$T(1) = 1$
3	$\sqrt{\frac{25}{4}} - \frac{1}{2} = 2$	$T(2) = 3$
15	$\sqrt{\frac{121}{4}} - \frac{1}{2} = 5$	$T(5) = 15$

Table 1.1

1.9 Remark — Translations Swap Under Inversion

The general pattern, stated cleanly: if you take the graph of f and translate it by the vector (h, v) in the plane, the graph of the inverse — which is the original graph reflected across $y = x$ — gets translated by the vector (v, h) . The components *swap*.

That is exactly what made this problem clean. T was the graph of S translated by $(-\frac{1}{2}, -\frac{1}{8})$, so T^{-1} is the graph of S^{-1} translated by $(-\frac{1}{8}, -\frac{1}{2})$ — and combined with $S^{-1}(y) = \sqrt{2y}$, the formula falls out by inspection. The square-root-of-something-plus-

stuff structure is no longer mysterious; it is the geometric shadow of " $\mathcal{S}^{-1}$ shifted left by $\frac{1}{8}$ and down by $\frac{1}{2}$."